

Meredith Huang

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EDUCATION

National Yang Ming Chiao Tung University

Bachelor of Electronics and Electrical Engineering

Sep 2022 – Jun 2026

Hsinchu, Taiwan

Coursework: Generative Artificial Intelligence, Machine Learning, Algorithms, Data Structures, Probability, Linear Algebra

University of Illinois Urbana-Champaign

Exchange Student in Electrical Engineering

Aug 2025 – Dec 2025

IL, US

Coursework: Applied Parallel Programming

SKILLS

Programming: Python, C++, CUDA, Golang, Git, Linux

Frameworks: PyTorch, TensorFlow, Scikit-Learn, NumPy, Diffusers, LangChain, FAISS, sentence-transformers

Cloud/DevOps: Azure, Kubernetes (K8s), Docker, Terraform

Tech/Fields: Computer Vision, Diffusion Models, Adversarial ML, 3D Generation, Generative AI, GPU Optimization

Languages: Mandarin (Native), English (IELTS 7.5)

PUBLICATIONS

Meredith Huang, Ling Lo, Tzu-Ling Lin, and Hong-Han Shuai, “A Blind Spot in Image Immunization: Subject-Preserving Background Replacement Reveals a Hidden Vulnerability,” *under review at AAAI 2027*.

- Proposed subject-preserving background replacement as an overlooked threat to diffusion image immunization.
- Designed reference-free trajectory and attention objectives requiring no clean image or defense knowledge.
- Benchmarked six defenses; proposed background-aware immunization cutting attack success from 96% to 10.5%.

RESEARCH & INDUSTRY EXPERIENCE

Research Intern, Lab for Digital Twin based Optimization, Delta Research Center

Jul 2026 – Aug 2026

- Built an end-to-end single-image 3D asset generation pipeline; benchmarked TRELLIS and Hunyuan3D.
- Fixed product-specific defects via guidance, LoRA, logo restoration, mesh decimation, and texture refinement.
- Trained a pairwise Bradley-Terry ranker to select the best of 12 candidates; VLM judges proved unreliable.

Research Assistant, Big Data Analytics and Social Intelligent Computing Lab

Feb 2026 – Present

Advisor: Prof. Hong-Han Shuai

Investigating adversarial machine learning methods for image and video immunization, focusing on evaluating defenses against Stable Diffusion editing and LoRA fine-tuning attacks.

Undergraduate Researcher, Mixed-Signal Electronic Design Automation Lab

Sep 2024 – Sep 2025

Advisor: Prof. Chien-Nan Liu

Explored ML-driven optimization for synthesis and floorplanning, focusing on constrained and timing-aware objectives.

PROJECTS

Document QA based on RAG — Python, FastAPI, LangChain, FAISS, BM25, Kubernetes, Docker, Terraform, Azure

- Built a RAG system for question answering over NLP papers with hybrid retrieval, HyDE, RRF fusion, and reranking.
- Deployed a lightweight cloud-native service on AKS using Docker/Kubernetes with Terraform-managed infrastructure.
- Improved answer correctness to 66/100 and evidence score to 0.35, outperforming baselines at 48/100 and 0.26.

Node Classification with GATv2 — Python, PyTorch, GATv2, GNN

- Built and tuned a GATv2 pipeline for multi-class node classification on a social graph.
- Improved accuracy by 4.2 points over the baseline using noise injection, mixed-precision training, and regularization.

GPU Optimization for CNN Forward Convolution — CUDA C/C++, GPU Profiling

- Optimized CNN forward convolution using im2col, kernel fusion, tiling, loop unrolling, and stream parallelism.
- Reduced end-to-end runtime by 47.9%, from 73.85 ms to 38.49 ms.

COMPETITION EXPERIENCE

ICCAD 2025 CAD Contest, Problem C by NVIDIA

Jan 2025 – Sep 2025

- Ranked 8th worldwide with Honorable Mention, improving TNS by 67.78% while slightly reducing HPWL and power.
- Built an end-to-end placement pipeline with legalization, detailed placement, STA feedback, and CUDA evaluation.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

PR Team Member, 11th Government X Meichu Hackathon

Feb 2022 – Nov 2023

Handled corporate outreach and sponsorship coordination with Google, 104 Job Bank, and Logitech.

Director of the HR Department, NYCU EE Student Association

Jun 2023 – Jun 2024

Managed recruitment, training, and staffing coordination for student association activities.